Exploring the Beauty Industry Through Cosmetic Network Analysis

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Abstract

This project applies network science to analyze the beauty industry through cosmetic formulation and the creation of two cosmetic networks: an ingredient co-occurrence network and a product-product network. Using a dataset of cosmetic products from a variety of online beauty retailers, the ingredient lists for each product were cleaned and then analyzed using Python scripts and Gephi for network visualization. Through use of centrality measures and community detection, the most common ingredients, compounds, and clusters of similar products were observed. The results showed that the ingredients of silica, mica, phenoxyethanol and dimethicone were the most common in the formulation of cosmetic products, and that lip and eye products shared a very similar formulation. Additionally, ingredients with potential risks such as parabens and heavy metals clustered together in the ingredient network. Further, the perspective of network science approaches onto cosmetic networks revealed formulation similarities across products and brands, along with patterns in ingredient pairings.

1 Introduction

1.1 Background

The beauty industry is one of the largest global markets, with an estimated $570 billion dollars in worldwide revenue last year [6]. Many people wear makeup on a daily basis, and continuously repurchase products, or use makeup as an outlet for their creativity. Therefore, it is vital to recognize that not every user has the same needs or wants in cosmetic formulation, as some may have oily versus dry skin and a variety of other needs. This creates an expansion of a standard formulation for cosmetics, and allows the market to serve to a variety of consumers of different skin types and skin colors. The standard labeling and advertising of products informs users which one would be better suited for there skin or preference. Through the rise of cosmetic diversity and demand, brands have consistently expanded and marketed different formulations in their cosmetics.

The biggest drawback to having a large variety of products, is that consumers are often overwhelmed with the amount of options provided, and they do not have the time or energy to check every ingredient in the cosmetics. This leads to a large gap into knowing what chemicals are being applied to one's skin on a daily basis. It is important to recognize that not all chemicals or chemical combinations will improve or be safe for people to wear regularly. This divide, creates an area of interest I wanted to explore. The chemical formulation for cosmetics is what drives teams behind makeup companies to constantly expand their line of products, advertising them as the "better" or improved versions of previous products they have launched. Specifically, there has been large efforts placed on clean beauty, with more natural ingredients and encouraging the use of safer ingredients in cosmetic products [2].

This project focuses on using network science on cosmetic product data by creating two networks, an ingredient co-occurrence network and a product-product network. The ingredient co-occurrence network observes the relationship between ingredients in cosmetics through the co-occurrence found in products, and the product-product network aims to analyze similarities in formulation across cosmetic categories and different brands. My main research question were: Which ingredients are the most common in cosmetics? Which ingredients are anchors in connections between different cosmetic categories? How similar are products in different brands and product lines? Do ingredients and products cluster into communities corresponding to specific uses or cosmetic categories?

1.2 Literature Review

1.2.1 The Flavor Network

The "Flavor Network and the Principles of Food Pairing" network analysis by Ahn et al.[1], was the initial motivation for this project. In their study, their network's goal was to observe the relationships between ingredient pairings accross different cuisines globally in different regions such as Western region and East Asia. Their networks consisted of ingredients as nodes and an edge connecting nodes weighted by their flavor compounds they found. In their study they found differences in ingredient pairings in different regional cuisines.

The Flavor Network inspired the use of cosmetic ingredient analysis, as both focus on ingredient and compound pairings and their co-occurrences. In cuisine, ingredient pairings change the flavor compounds found and form patterns in North America and East Asian cuisine, while the cosmetic ingredient pairings reveal the cosmetic formulation used for making products adjust texture and pigment based on the goals of the product line. The studies use of weighted co-occurrence ingredient pairings served as a basis for the first network in this project.

Though, an emphasis must be made on contextual analysis is different in cuisine ingredient pairings and cosmetic ingredient pairings, as the flavor network focuses on seeing the flavor compounds in cuisines which varies widely on preferences of people in those regions, meanwhile, in the cosmetic ingredient network, a lot of the pairings are derived from standard formulations needed to maintain product quality or needs.

1.2.2 Cosmetic Recommendation Systems

There has been an increase in using deep learning methods and artificial intelligence for personalized cosmetic recommendation systems, specifically in skincare. In a study by Lee et al [7], they created a deep learning-based cosmetic recommendation system which combined AI for skin analysis of individuals and analyses of cosmetic ingredients. Their emphasis on ingredients allows for an extensive review of which cosmetics will improve or help cater to a customer's skin needs, rather than providing a recommendation based on product branding. Previous AI use in cosmetic recommendation systems have either focused on using only user reviews or not performing actual skin analysis of individuals, creating a gap in personalized recommendations [7].

Through their system, they used Transformer-based neural networks to analyze ingredients in products along with U-Net based segmentation networks to identify skin conditions in individuals such as acne. Their analyses highlights the importance of prioritizing ingredient and personalize skincare recommendations in the cosmetic industry. Their approach reflects the trend of the cosmetic industry starting to shift towards clean beauty and incorporating less harmful ingredients in their products, to prioritize customers' skin health [2].

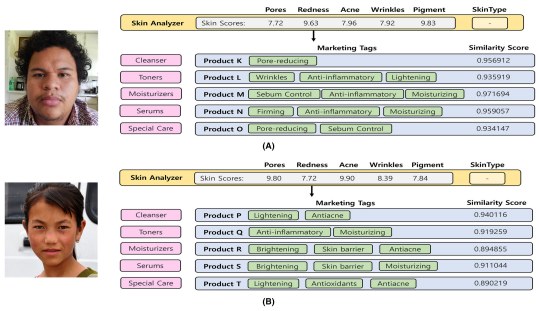


Figure 1: Recommendation results of a public dataset, demonstrating the skin analysis on two users and personalized skincare recommendations.

The study provided a structured approach using deep learning methods to predict personalized product recommendations, instead of focusing on the relationship between particular cosmetic formulations. Therefore, taking a network science approach would differ in examining the patterns of ingredient co-occurrence in products, and similarity between cosmetics, along with examining which ingredients dominate cosmetic product formulation.

2 Methods

2.1 Data

The data used for analysis in this project was found on Kaggle, titled "E-commerce Cosmetic Products" [5]. The dataset contained over 110,000 cosmetic products gathered from online retailers including Amazon, Sephora, Ulta, and Flipkart. Each product had information on product name, brand, category, subcategory, price, rating, and ingredient lists.

In this analysis, only products in the "eyes", "lips", and "face" categories were filtered to, to focus on makeup products such as lipsticks, concealers, blush, mascaras, eyeliners, and eyeshadows which are common face cosmetics. Any products that did not include ingredient lists were filtered out from the dataset.

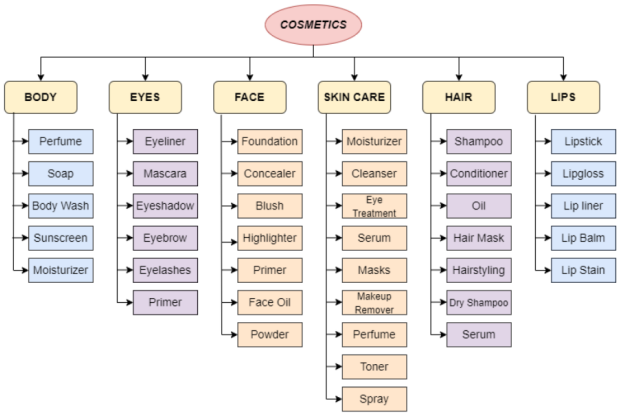


Figure 2: Categories Provided in the original Kaggle Dataset

Further, the ingredient lists required preprocessing before the network could be created. The ingredient list strings were converted to all lowercase and cleaned by removing any numbers, percentages, punctuation, and extra white space in the string by using Python. Then, the ingredient lists were separated at the commas, to isolate individual ingredients and place them into sets of unique ingredients for each cosmetic products.

2.2 Ingredient Co-Occurrence Network

The ingredient co-occurrence network was created by representing cosmetic ingredients as nodes. Then, an undirected edge would be created between two ingredients whenever they both appeared together in the same product. The edge weights correspond to the number of products in which the unique pair of ingredients had co-occurred. In each product, the pairwise ingredient combinations were made from using Python's combination function. Then, repeated ingredient pairs across products would increase the edge weight, which allowed for observing ingredient pairings across products and their frequency. Finally, the node and edge lists were exported as CSV files which were then exported from the google Colab notebook to use for visualization and further community analysis using Gephi. Further, backbone extraction was then used to filter out the network's edge to reduce density, and the nodes were filtered to only include ingredients nodes with a weighted degree of 95 and above, reducing the nodes to include only 33.58% of the original network nodes.

2.3 Product-Product Network

The product-product network was created to represent cosmetic products as nodes and their edges corresponds to shared ingredients in their formulations. Two products in the network were connected by an undirected edge if they shared a minimum of 5 shared ingredients. The shared ingredients were placed as the edge weights, to observe the similarity between products formulations.

For accessible visualization, to reduce the density of the network, the edges were filtered to improve visibility. The lower the edge threshold (number of shared ingredients between two products), the denser the network became, with the higher thresholds provided clearer visibility. After testing several thresholds, the final filter was placed on a minimum of 13 shared ingredients between products as this preserved the core structure of community connections and reduced the noise on the outside parts of the network.

Several centrality measures were computed for both networks to analyze ingredient and product similarities. The centrality measures included betweenness centrality, eigenvector centrality, and PageRank centrality. Additionally, the weighted degree of nodes was used to identify which ingredients were the most common in the cosmetic products, and in the product network to see which products had the highest number of shared ingredients. The betweenness centrality was used to identify ingredients that acted as connectors throughout several communities, while the PageRank was used as a measure of which ingredients had the most influence within the cosmetic network. Then, modularity was ran to identify communities within both networks in Gephi. The node sizes and their labels were sized according to the highest 12 ranked nodes based on betweeness centrality to identify the vital ingredients and products in the networks.

2.4 Trial and Error

There were several struggles in the initial network construction process as the ingredient listed included many ingredients that were the same, but only different in concentration represented by the differences in $4.5\%, 7.5\%$ and others, requiring a cleaning of the ingredient listing. This was the main concern in the beginning, as it required other re-formatting to the ingredient listings to ensure there were not duplications within the network that would provide in inaccurate analysis or communities clustering.

Another challenge was the density of the ingredient co-occurrence network, as the ingredient co-occurrences were so large that the network was very dense and clustered. Leaving no option, but to resort to back boning the network to keep the significant structure of it, and improving network visuals. After back boning was done, filtering by weighted degree was required as there were too many nodes for useful interpretation, with some ingredient pairings being very low in frequency such as 10 or 20 times only, therefore finding the right threshold for filtering nodes was important. After testing several, I found that filtering to nodes with a weighted degree of 95 and above maintained the core structure of the network while still being able to interpret. The average weighted degree of 95 represents the minimum number of co-occurrences for ingredient pairings. Further, there were many coding challenges present when attempting to create both networks as many factors including assuring unique ingredient pairings and increasing edge weights with co-occurrences and product shared ingredient edge weights, were not intuitive. To help with this challenge, Claude AI was used to supplement and de-bug code in Python, assisting with structuring proper for loops and setting edge weights for the networks.

The final challenge in this project was classifying communities in both networks, as the highest ranked nodes in communities based on betweenness centrality were sometimes not within a cohesive theme of ingredients within their community, or products in communities were not all from the same categories or brands. To address this, I manually copied and pasted the ingredients into google's search engine and prompted it to help classify the use of the majority of the ingredients in the community, and identify what these ingredients represent in terms of classifying them as texturizers, pigments, emulsifiers and more. For the product network, I took a simpler approach, and mainly focused on the category and subcategory of the products within the communities, attempting to observe if there was a cohesive theme of product categories in the major communities or if they were more diverse amount of categories present. This took careful consideration in classifying communities, as there was a concern in missing some of the less prevalent (by centrality measure) ingredients or products, and prioritizing which the major structure of each community niche within the network.

3 Results

Analyzing both the ingredient co-occurrence and the product-product network were the focus of investigating the beauty industry through cosmetic networks. They both provided reliable insights on chemical formulation in cosmetics and the similarities between cosmetic categories and different brands. Through centrality measures and community identification it demonstrated groups of ingredient and product clustering in brands and categories. The following sections interpret results from communities groupings and the measures ran on both networks in the

cosmetic context.

3.1 Ingredient Co-Occurrence Network

The ingredient co-occurrence network displayed highly dense and interconnected structure of ingredients that frequently appeared together in cosmetic products. After backbone extraction and filtering the nodes in Gephi, the network demonstrated a compact and dense core structure with multiple communities with ingredients related to one another within each group.

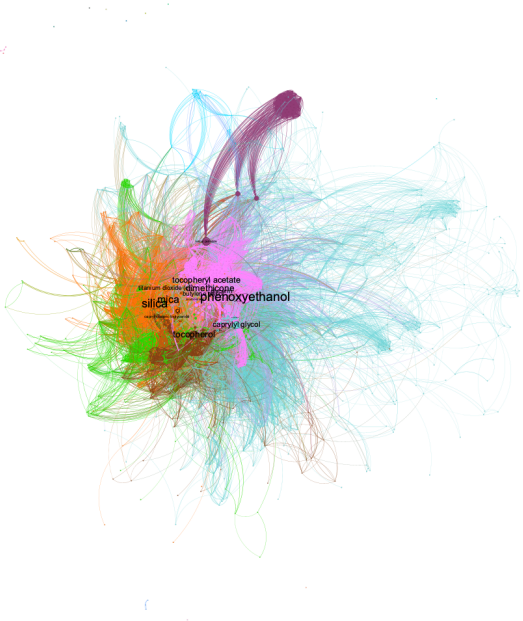


Figure 3: Ingredient Co-Occurrence Network Filtered with Backbone Extraction and Weighted Degree of above 95

Gephi identified 26 communities in the filtered network. The largest communities displayed in Figure 3, appear to be highly interconnected and there are no noticeable disconnected components present in the network. Reflecting the common formulation of products which are made up of different ingredients that all need preservatives and pigments among other ingredients, that are commonly used in any cosmetic product. Analyzing the ingredients in the largest communities:

- Orange Community: Most of this community was dominated by pigments and fillers, and also contained many of the most common ingredients in the network. Including: silica, mica, and titanium dioxide (ci).
- Pink Community: Another highly connected community of the most common ingredients across products including: phenoxythenol, dimethicone, butylene glycol, glycerin, tocopheryl acetate. Making this community contain many of the silicone ingredients along with ingredients meant to extend the product's wear, as phenoxythenol is used as a preservative across most makeup products [4]. Meanwhile dimethicone is typically used for texture purposes in products to provide a smooth and consistent application of makeup.
- Blue Community: This community contained mainly emulsifiers and ingredients typically found in cream based cosmetic products such as skincare products. The most common ingredients in this group were caprylyl glycol, polysorbate, pentylene glycol, and glyceryl stearate. These compounds are important in acting as stabilizer for the oil water mixtures in products and keeping the formulation consistent in liquid or cream products [9]. This suggests that different cosmetics share similar emulsifying systems for application regardless of the category of the product or brand.
- Green Community: Consisted of many waxes, natural butters, water and emsulsifiers. Some of the top betweeness centrality ingredients were butyrospermum parkii (shea) butter, water, glycine soja (soybean), and parfum/fragrance.

- Brown Community: There are many fragrance allergens present in this group, such as limolene and citral, along with waxes of candelilla wax and rice bran wax, with the third dominant ingredients being clay minerals like aluminum hydroxide and kaolin very prevalent in more natural beauty [2].
- Violet Community (top middle): One of the most suprising findings in the ingredient network was this group of the commonly restricted or concerning and toxic chemical ingredients found in cosmetic products. Ingredients in this community were compounds like parabens, lead, mercury, pfas, and oxybenzone, talc, and retinyl palmite (labeled on the network). Though some of these ingredients can be used in cosmetics for pigmentation, some of the ingredients have become safety concerns as they have links to health risks [10]. Therefore this clustering of risky ingredients suggest that these ingredients tend to co-occur in particular formulation or specific product. Further, the fact that this group is mainly connected to the rest of the network by only a few anchor nodes such as retinyl palmite is a good sign that they are not common amongst most products in the dataset, and only in certain products. Highlighting that there is a group of concerning ingredients in some product formulation, and that customers should take notice of these often restricted compounds or ingredients for skin safety and health in general.

3.1.1 Observing Centrality Measures in Ingredient Co-Occurrence Network

Focusing on centrality measures further demonstrated which ingredients are the most common in cosmetic formulation. As the nodes represented the ingredients, the higher the degree of a node represent a more commonly used ingredient in cosmetic products. Using Gephi to gather the highest ranked nodes in weighted degree, it identified the most common ingredients.

Table 1: Top 10 Ingredients by Weighted Degree (Most Co-occurring Ingredients)

Rank	Ingredient	Weighted Degree	PageRank	Community
1	silica	30,283	0.0140	0
2	mica	29,758	0.0136	0
3	phenoxyethanol	28,449	0.0135	1
4	titanium dioxide (ci)	25,819	0.0116	0
5	dimethicone	25,400	0.0117	2
6	tocopherol	23,928	0.0112	3
7	caprylyl glycol	23,274	0.0109	1
8	tocopheryl acetate	22,943	0.0107	2
9	ci	22,638	0.0103	1
10	disteardimonium hectorite	18,401	0.0082	2

Through the weighted degree analysis shown in Table 1, it was observed that silica, mica, pheoxyethanol, titanium dioxide (ci), and dimethicone were the most highly connected ingredients within the network. These ingredients appear across a variety of the cosmetic categories and communities as they all serve different vital properties in cosmetics such as the pigmentation, texture improvement, and preservation importance in formulation [8].

Table 2: Top 10 Ingredients by Betweenness Centrality

Rank	Ingredient	Betweenness Centrality	Community
1	phenoxyethanol	0.0306	1
2	silica	0.0276	0
3	tocopherol	0.0241	3
4	mica	0.0241	0
5	dimethicone	0.0234	2
6	tocopheryl acetate	0.0229	2
7	retinyl palmitate	0.0224	4
8	caprylyl glycol	0.0206	1
9	butylene glycol	0.0195	1
10	glycerin	0.0193	1

The betweenness centrality highlighted ingredients that serve as connectors between different ingredient communities. In Table 2, observe that phenoxyethanol had the highest betweenness centrality, indicating that it's common use as a preservative in makeup appears regardless of product type or brand. This is likely due to makeup needing to be shelf life stable for months if not a year after the cosmetic product is produced, and therefore it is vital to include this ingredient in any cosmetic product [4]. Similarly, the ingredients of glycerin, butylene, and caprylyl glycol are very connected in formulation focused on hydration properties most often found in silicone based products. Therefore, these ingredients connected the different cosmetic products to have similar purposes in terms of being used as primers and foundations despite different brands producing them with different formulations. Further, viewing retinyl palmitate and tocopherol in the top 10 ranking of betweenness centralities reflect the common overlap in cosmetic products of foundation/primers/concealers and skincare products, as the shift towards simple, natural, and clean ingredients is become popular in the cosmetic industry, therefore they are more likely to share formulation.

3.2 Product-Product Network

The ingredient network prioritized observing the relationships between different ingredients and chemical compounds, meanwhile the product-product network focuses on the relationships between cosmetic products through their shared ingredients. In this network, as the nodes represent products and the edges connect the product that share a certain number of ingredients, there were several similarities observed. Previously mentioned, the network contains products that share at least 13 ingredients and this narrowed down a nice visual representation of the network structure which contained 2,787 nodes and 32,080 edges, with the number of filtered edges being approximately 4.13% of the original network.

The filtered network demonstrated a core like structure in the center with many interconnected communities which were surrounded by smaller clusters on the outskirts of the network. Through community detection in Gephi, it identified 977 communities in total, noting the diverse quantity of formulations that have similar formulation in cosmetic products and brands.

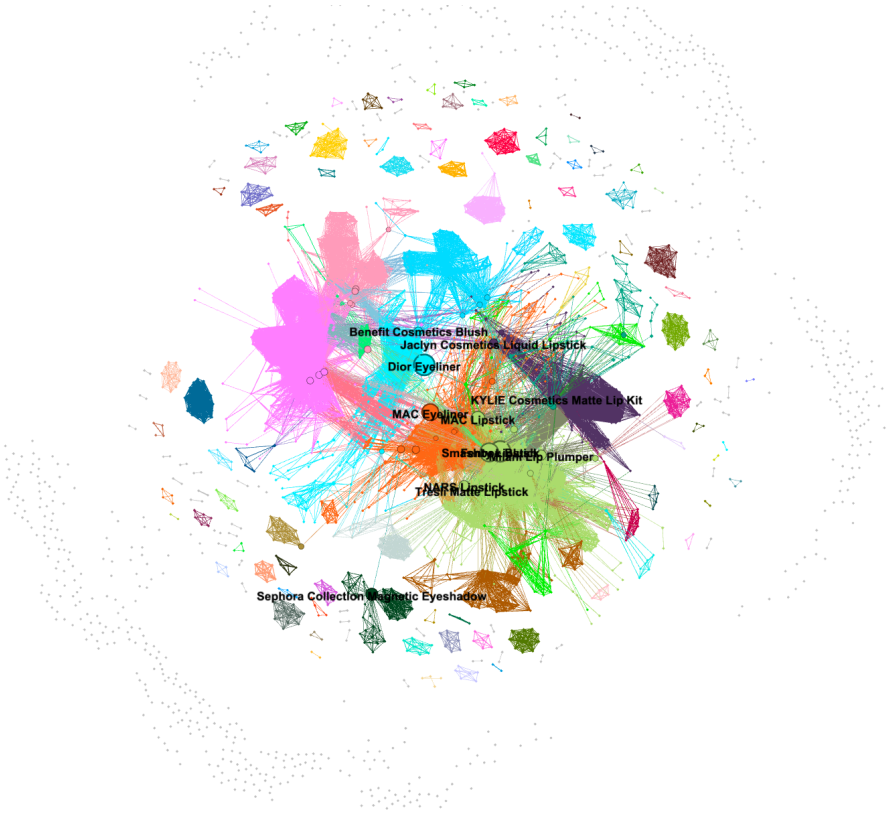


Figure 4: Product-Product Network of 13 or more Shared Ingredients between products

Analyzing the products and their respective categories in the communities displayed in the Figure 4 network:

- Dark Pink Community (Large on the left of the core): One of the largest communities in the network contains only lipstick and lip gloss products, highlighting the strong base formulation shared across these two categories which were domiated by waxes, pigments, and oils most commonly used in lip product formulations.
- Light Blue Community (Large center of core): Contains mainly eye products:eyeliners, mascara, eyebrow with a few quantities of concealers and blush from Benefit Cosmetics, Sephora Collection, and Makeup For Ever Brands mainly.
- Green Community (Large lower left of the core): Consisted of only lipsticks, lip glosses, and lipbalms from a variety of brands inclduing: Huda Beauty, Clinique, Fenty Beauty, and Smashbox.
- Orange Community (Large lower center of core): Mainly eye products: eyeliners, eyeshadows, eyebrows with some scattered products of lipsticks and blush from a large variety of brands.
- Light Pink Community (Upper left of core): Mainly concealers, with a tiny portion including blush products from the same brand of concealers (Makeup For Ever Brand, Sephora Collection Brand, and Givenchy Brand).
- Light Pink Community Cluster (Above the Blue Community): This smaller community demonstrated brand specific clustering. This group had only HUDA Beauty Brand concealers, with different shade variations or very minor differences in concealer type.
- Dark Green Community (Below the large core): Selective community of only Sephora Collection Brand Metallic eyeshadow and eyeliners.

Noting from Figure 4, that the nodes(products) with the largest betweenness centralities lie in the large communities of blue, orange, and light green. After observing the largest communities cosmetic categories and noticing that some of them had a mixture of eye products and lip products, I wondered what the smaller clustered and disconnected communities could consist of. My hypothesis on these small communities were that they were very unique products from the same line of a brand. After reviewing a handful of these communities, such as the light pink community above the core cluster, it was validated. This community included only Huda Beauty concealers, with no other category or brands in this cluster. Reflecting that many brands keep a consistent formulation for product types and that these small clusters reflect how cosmetic lines are constructed around standard formulations that are only modified by a few ingredients between brands.

Further, I hypothesized that many of the disconnected nodes on the very peripheral of the network are very unique products with ingredients that are not commonly used, therefore not forming connections with any other communities or clusters in the network. These products containing uncommon ingredient combinations or for specific uses such as metallic eyeshadow reflect products that form part of specific advertising for special use or clean beauty brands that stick to a unique combination of ingredients and they do not stray too far from their base formulation across their different product lines.

3.2.1 Observing Centrality Measures of Product Network

Table 3: Top 10 Products by Betweenness Centrality

Rank	Product	Betweenness Centrality	Category
1	Dior:Diorshow 24H* Stylo Eye Liner - 061 Matte Grey	0.0306	Eyes
2	FENTY BEAUTY by Rihanna:Fenty Icon Semi-Matte Refillable Lipstick	0.0276	Lips
3	Smashbox:Starlaa Rosy Bronze Blush	0.0241	Face
4	MAC:Pro Longwear Fluidline	0.0241	Eyes
5	MAC:Locked Kiss Ink Lipstick	0.0234	Lips
6	Sephora Collection:Colorful Magnetic Eyeshadow - 29 First Light	0.0229	Eyes
7	Milani:Keep It Full Maxxx Lip Plumper	0.0224	Lips
8	NARS:Afterglow Sensual Shine Lipstick	0.0206	Lips
9	Jaclyn Cosmetics:Strawberry Feels Poutspoken Liquid Lipstick	0.0195	Lips
10	Benefit Cosmetics:Dandelion Baby Pink Brightening Blush Travel Size Mini	0.0193	Face

The betweenness centrality in the product network identified the products that serve as connectors between communities. From Table 3 results show that products such as Dior eyeliner, Fenty Beauty lipsticks, Mac eyeliner, and Smashbox blushes occupied the important positions connecting different product clusters. Therefore, these products most likely contain ingredient combination that are used in a variety of product types and brands, bringing together the communities of lip, eye, and face products. The most notable observation was the strong presence of lip and eye product formulation similarities. After investigating the correlation, it appears that lipsticks and eye products have very similar formulation because they both require the products to be hydrating and gentle ingredients due to the eye and lip regions being prone to dryness and sensitivity.

Table 4: Top 10 Products By PageRank Centrality

Rank	Product	PageRank Centrality	Category
1	Smashbox:Starlaa Rosy Bronze Blush	0.0306	Eyes
2	Clinique:Even Better Pop Lip Colour Foundation - Blush	0.0276	Lips
3	Clinique:Even Better Pop Lip Colour Foundation - Nuzzle	0.0241	Face
4	Clinique:Even Better Pop Lip Colour Foundation - Subtle	0.0241	Eyes
5	Clinique:Even Better Pop Lip Colour Foundation Lipstick	0.0234	Lips
6	Clinique:Even Better Pop Lip Colour Foundation - Enamored	0.0229	Eyes
7	Clinique:Even Better Pop Lip Colour Foundation - Nestled	0.0224	Lips
8	Clinique:Even Better Pop Lip Colour Foundation - Closer	0.0206	Lips
9	Clinique:Even Better Pop Lip Colour Foundation - Cuddle	0.0195	Lips
10	Clinique:Even Better Pop Lip Colour Foundation - Tender	0.0193	Face

Then observing the PageRank centralities of the products, it revealed the most prevalent products in the network in terms of most similar ingredients. In Table 4, it can be observed that Clinique products dominated the highest ranked by PageRank products. This finding suggests that the Clinique formulation share a majority of the same formulation structure with other cosmetic products across the network. Therefore, instead of having very distinct or unique formulation, these Clinique products seem to have central roles within the cosmetic ingredient formulation.

3.3 What Makes Clinique Special?

Observing the dominance of Clinique products in the PageRank analysis of the product network demonstrated a trend in their product branding and cosmetic formulation. Clinique is a beauty brand based on providing customers with clean more skincare based focused, dermatologist tested, and non-toxic ingredients with simple formulation in

their products [3]. Therefore, many of their products depend on base cosmetic ingredients shared in formulation across the cosmetic industry. These include similar ingredients of moisturizers, pigments, and silicone ingredients highly used in products that mimic skincare like and more natural makeup.

Therefore, Clinique products are central to the network because they provide a base formulation that reflects other formulations in the networks across different product types. This result suggests that products from brands that are branded as clean beauty do not use specific uncommon ingredients, and rather rely on a combination of commonly used cosmetic ingredients and compounds. Further, the findings indicate that many other cosmetic brands may chose to build on these base formulation for their products to modify for skin types, shades, and other attempts to personalize cosmetic products to customers. This was the most intriguing finding of the network, and emphasizes the importance of using natural or safer ingredients in products for better skin health. Many brands have adopted this motto of safer and cleaner cosmetics, to appeal to customers searching for natural ingredients and cosmetics.

Both networks revealed insights on the structure of the cosmetic industry which is dominated by a particular set of reoccurring ingredients and formulations. Though the cosmetic products can be branded as being as personalized to customers in the general consumer, the network demonstrated that many products have similar ingredient composition across the brands and cosmetic categories. The main core of ingredients included those used for preservative, pigments, and silicone in the cosmetic formulations and reflect the base formulation in products.

4 Discussion

Throughout this project, the most important findings was the main structure in cosmetic formulations. Though there appeared to be some diversity within the different cosmetic brands their types of products, the first network of the ingredient co-occurrence showed that there are specific ingredients that are very commonly used in most cosmetic products. These ingredients included silica, mica, dimethicone, glycerin, and phenoxyethenal that all stayed in the top ranking across centrality and weighted degree measures, as they consistently appeared across a variety of categories and products.

The finding suggests that a majority of cosmetic products are built from a base formulation and are then adjusted or more ingredients are added to it, for specific purposes of shade matching, or texture preferences. Though many brands may advertise their products as revolutionary and different than their competitors, the network showed that the same most common ingredients are typically found in these products. Therefore, products may not differ more strongly in ingredients, but rather in other aspects of the product such as the brand or their target audience and not so much a totally different formulation than their competitors.

Then, the product-product network revealed hidden similarities between cosmetic products. The highly connected clustering between lipsticks, eyeliners, mascaras and other face products showed that though these products may be cross categorical, they have many shared ingredients in their formulations. Some of the groupings also demonstrated specific clustering of brands and product lines within those particular brands with only slight shade or small differences between these products.

These analyses highlight the difference in product marketing and the actual formulation of products. The cosmetic industry brands their products as being often unique and helpful for specific consumer preferences or needs, yet the networks showed that many products share similarities in formulation. Though this result does not mean that all formulations are going to be the same or majority of the same ingredients, it does show that cosmetic product advertised as unique may be more similar than expected. Through these networks, it provided insight on the cosmetic formulation beyond just the branding label of products.

Connecting one of the most important finding of unregulated ingredients in the ingredient network, it highlights the importance of brands being consumers friendly and upfront on the formulation of products. In the ingredient community of these commonly unsafe ingredient of oxybenzone, talc, and other heavy metals in the network community it demonstrated how a network science approach to cosmetic can identify a group of ingredients typically associated with harmful health concerns. Though these ingredients presence in the network was more separated to the rest of the communities, their clustering being in the same community suggested that they are typically combined in products, therefore some products may be more harmful than others. In particular, the more metallic

and pigmented eyeshadow and lipsticks that are susceptible to including more of those heavy metals to get the shine and pigment that consumers want. It is vital for brands to be transparent on the risks of the ingredients being used in the product, for the safety of customers, and for them to make an informed decision before buying these products.

The lack of transparency many brands failed to provide, resulted in the shift currently observed in the cosmetic industry of cleaner and safer makeup brands. Therefore ingredient analysis is key to making the transition from harmful ingredients, to safer alternatives in products. By using network analysis, it helps observe the specific ingredient connections within a product, rather than the overarching formulation of a product as it becomes increasingly difficult to analyze 10+ ingredients contained in a singular product. These trend analyses in ingredients and within products will help brands become safer and expand into the cleaner beauty cosmetic branch.

4.1 Limitations and Further Analysis

There were several limitations in the analysis of both networks. To begin, the dataset used in this project relied on ingredient listing from online retailers that may not be up to date with their current formulation of products for brands as the dataset was last updated two years ago in 2024, as many brands may update their formulas for an improved version of their products. Additionally, the ingredient list only provides the number of ingredients within the product and not the exact amounts present in each. Therefore no analysis could be made on the most prevalent ingredients within a product. Leaving the ingredient co-occurrences to be classified the same, rather than on a quantity of each ingredient basis in the products.

Further, the product-product network was filtered down by edges. Therefore, a different filtering of edges would result in different community structures depending on how dense or relatively sparse the network would become if you increased or decreased the threshold of shared ingredients between products. Another limitation was that pricing, ranking, nor customers reviews were incorporated into the analysis, which could have provided an additional perspective to the efficiency of certain cosmetic products and if there a difference in product formulation for products in lower-end brands versus higher-end brands, for example whether certain ingredient compounds are found in more popular or "better" cosmetic products.

Further work would include incorporating datasets that rank ingredient safety in cosmetics, or the toxicity of the ingredients to analyze whether certain products exhibit more at-risk ingredients. Another extension would be creating a bipartite network similar to the one introduced in the Flavor Network analysis, but in this context would connect products with ingredients. This could provide insight on which ingredient compounds are most prevalent in certain product types. Along with potentially gathering another dataset of cosmetic ingredient data in the past, to compare how brands have updated their formulations to see if they have included certain ingredients more often now, and what trend of ingredients have increased across products. The last potential extension would be to include other products in the beauty industry, and not just cosmetic products. The dataset has information on haircare and fragrance products amongst other, therefore analysis could be made to observe if any patterns in ingredient combination extend beyond cosmetic product formulation and into other types of beauty products.

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5 Appendix

For code in constructing the network's review the zip file titled "DomCapstoneFiles". Inside the folder there is the Python script used to create the networks titled "DominguezCosmeticNetworks(1).ipynb". Once the script is ran it exported a node and edge csv files for the ingredient co-occurence network (this is titled "ingredient-nodes (1).csv" and "updated-ingredient-edges.csv" in the folder, import these two into Gephi as undirected graph) along with a node and edge csv files for the product-product network (this is titled "product-nodes.csv" and "product-edges.csv" in the folder, import these into Gephi as undirected graph). Additionally, in the folder, is the datatset used in this project titled "real-cosmetics.csv", along with the Gephi exported workspaces of the networks.